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## Mathematical model

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$P(t)$  is the number of rainbowfish in the aquarium, with  $t$  in days.

The birth rate of the fish is  $b = 0.7$  per day. The death rate is ignored:  $d \approx 0$ .

Now, 20 fish per day are sold.

In the exercises, you will derive the second mathematical model.

### Exercise A

1/1 point (graded)

Derive a new balance equation, where  $\Delta t$  is the length of a time interval, and  $\Delta P$  is the change in  $P$  during this time interval.

The balance equation now becomes

☐  $\Delta P = -20\Delta t$

☐  $\Delta P = 0.7P(t)\Delta t - 20$

☒  $\Delta P = 0.7P(t)\Delta t - 20\Delta t$  ✓

☐  $\Delta P = 0.7P(t) - 20$

Submit

You have used 1 of 2 attempts

**i** Answers are displayed within the problem

### Exercise B

1/1 point (graded)

Use the new balance equation to derive the new differential equation when 20 fish per day are sold. Note that you should enter your answer in the form  $a * P + b$

Type the right hand side of  $\frac{dP}{dt} =$

0.7\*P-20

✓ Answer: 0.7\*P-20

$0.7 \cdot P - 20$

Submit

You have used 1 of 3 attempts

**i** Answers are displayed within the problem

Now you have derived the differential equation for the problem, it is time to investigate the solutions in the next section.

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